

Java Date and Time

Problem | Submissions | Leaderboard | Discussions | Editorial

The `Calendar` class is an abstract class that provides methods for converting between a specific instant in time and a set of calendar fields such as `YEAR`, `MONTH`, `DAY_OF_MONTH`, `HOURL`, and so on, and for manipulating the calendar fields, such as getting the date of the next week.

You are given a date. You just need to write the method, `getDay`, which returns the day on that date. To simplify your task, we have provided a portion of the code in the editor.

Example

`month = 8`
`day = 14`
`year = 2017`

The method should return `MONDAY` as the day on that date.

AUGUST 2017						
SUN	MON	TUE	WED	THU	FRI	SAT
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

```
3 import java.security.*;
4 import java.text.*;
5 import java.util.*;
6 import java.util.concurrent.*;
7 import java.util.function.*;
8 import java.util.regex.*;
9 import java.util.stream.*;
10 import static java.util.stream.Collectors.joining;
11 import static java.util.stream.Collectors.toList;
```

```
12
13 class Result {
14
15     /*
16      * Complete the 'findDay' function below.
17      *
18      * The function is expected to return a STRING.
19      * The function accepts following parameters:
20      * 1. INTEGER month
21      * 2. INTEGER day
22      * 3. INTEGER year
23      */
24
25     public static String findDay(int month, int day, int year) {
26         Calendar cal = Calendar.getInstance();
27
28         // Calendar months are 0-indexed (January is 0), so subtract 1
29         cal.set(year, month - 1, day);
30
31         // Fetch the full name of the day in US English and capitalize it
32         return cal.getDisplayName(Calendar.DAY_OF_WEEK, Calendar.LONG, Locale.US).toUpperCase();
33     }
34 }
35 }
```

```
36
37 public class Solution {
38     public static void main(String[] args) throws IOException {
39         BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
```

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

- ✓ Test case 0
- ✓ Test case 1
- ✓ Test case 2
- ✓ Test case 3
- ✓ Test case 4
- ✓ Test case 5

Compiler Message

Success

Input (stdin)

1 08 05 2015

Download

Expected Output

1 WEDNESDAY

Download

1360. Number of Days Between Two Dates

Easy Topics Companies Hint

Write a program to count the number of days between two dates.

The two dates are given as strings, their format is `YYYY-MM-DD` as shown in the examples.

Example 1:

Input: date1 = "2019-06-29", date2 = "2019-06-30"
Output: 1

Example 2:

Input: date1 = "2020-01-15", date2 = "2019-12-31"
Output: 15

Constraints:

- The given dates are valid dates between the years 1971 and 2100.



Seen this question in a real interview before? 1/6

Yes No

Accepted 77,418/145.6K | Acceptance Rate 53.2%

Topics

443 42 9 Online

Solved

Code

Java Auto

```
21     for (int m = 1; m < month; m++) {
22         days += daysInMonth[m - 1];
23         if (m == 2 && isLeap(year)) {
24             days++;
25         }
26     }
27     days += day;
28
29     return days;
30 }
31
32 private boolean isLeap(int year) {
33     return (year % 400 == 0) || (year % 4 == 0 && year % 100 != 0);
34 }
35 }
36
```

Saved

Ln 36, Col 4

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

date1 =
"2019-06-29"

date2 =
"2019-06-30"

Output

1

Expected

1

DescriptionAccepted XEditorialSolutionsSubmissions

1154. Day of the Year

EasyTopicsCompaniesHint

Given a string `date` representing a [Gregorian calendar](#) date formatted as `YYYY-MM-DD`, return the day number of the year.

Example 1:

Input: `date = "2019-01-09"`
Output: 9
Explanation: Given date is the 9th day of the year in 2019.

Example 2:

Input: `date = "2019-02-10"`
Output: 41

Constraints:

- `date.length == 10`
- `date[4] == date[7] == '-'`, and all other `date[i]`'s are digits
- `date` represents a calendar date between Jan 1st, 1900 and Dec 31st, 2019.

UNLOCK EXPERIENCES

Prepare your network for today and tomorrow

HPE

Seen this question in a real interview before? 1/6

YesNo

Accepted 101,722 / 202.9K | Acceptance Rate 50.1%

5183512 Online

Code

JavaAuto

```
1 class Solution {
2     public int dayOfYear(String date) {
3         int year = Integer.parseInt(date.substring(0, 4));
4         int month = Integer.parseInt(date.substring(5, 7));
5         int day = Integer.parseInt(date.substring(8, 10));
6
7         int[] daysInMonth = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
8
9         int ans = day;
10
11         for (int i = 0; i < month - 1; i++) {
12             ans += daysInMonth[i];
13         }
14
15         // Add one day if it's a leap year and date is after February
16         if (month > 2 && isLeap(year)) {
17             ans++;
18         }
19     }
20 }
```

SavedLn 26, Col 2

TestcaseTest Result

AcceptedRuntime: 0 ms

Case 1Case 2

Input

date =
"2019-01-09"

Output

9

Expected

9

Contribute a testcase

1185. Day of the Week

Solved

Easy Topics Companies Hint

Given a date, return the corresponding day of the week for that date.

The input is given as three integers representing the `day`, `month` and `year` respectively.

Return the answer as one of the following values `["Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"]`.

Note: January 1, 1971 was a Friday.

Example 1:

Input: day = 31, month = 8, year = 2019

Output: "Saturday"

Example 2:

Input: day = 18, month = 7, year = 1999

Output: "Sunday"

Example 3:

Input: day = 15, month = 8, year = 1993

Output: "Sunday"

Constraints:

- The given dates are valid dates between the years 1971 and 2100.



461 44 9 Online

Code

Java Auto

```
1 class Solution {
2     public String dayOfTheWeek(int day, int month, int year) {
3         String[] week = {
4             "Friday", "Saturday", "Sunday",
5             "Monday", "Tuesday", "Wednesday", "Thursday"
6         };
7
8         int[] daysInMonth = {
9             31, 28, 31, 30, 31, 30,
10            31, 31, 30, 31, 30, 31
11        };
12
13        int days = 0;
14
15        // Add days for complete years
16        for (int y = 1971; y < year; y++) {
17            days += isLeapYear(y) ? 366 : 365;
18        }
19    }
20}
```

Saved

Ln 37, Col 2

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

day =
31

month =
8

year =
2019

Output

"Saturday"

Java Priority Queue

[Problem](#)[Submissions](#)[Leaderboard](#)[Discussions](#)[Editorial !\[\]\(2b376d1a92330ab09dad2665d2f89bf5_img.jpg\)](#)

In computer science, a priority queue is an abstract data type which is like a regular queue, but where additionally each element has a "priority" associated with it. In a priority queue, an element with high priority is served before an element with low priority. - [Wikipedia](#)

In this problem we will test your knowledge on [Java Priority Queue](#).

There are a number of students in a school who wait to be served. Two types of events, ENTER and SERVED, can take place which are described below.

- ENTER: A student with some priority enters the queue to be served.
- SERVED: The student with the highest priority is served (removed) from the queue.

A unique id is assigned to each student entering the queue. The queue serves the students based on the following criteria (priority criteria):

1. The student having the highest Cumulative Grade Point Average (CGPA) is served first.
2. Any students having the same CGPA will be served by name in ascending case-sensitive alphabetical order.
3. Any students having the same CGPA and name will be served in ascending order of the id.

Create the following two classes:

- The Student class should implement:
 - The constructor `Student(int id, String name, double cgpa)`.
 - The method `int getID()` to return the id of the student.
 - The method `String getName()` to return the name of the student.
 - The method `double getCGPA()` to return the CGPA of the student.
- The Priorities class should implement the method `List<Student> getStudents(List<String> events)` to process all the given events and return all the students yet to be served in the priority order.

Input Format

The first line contains an integer, n , describing the total number of events. Each of the n subsequent lines will be of the following two forms:

- ENTER name CGPA id: The student to be inserted into the priority queue.
- SERVED: The highest priority student in the queue was served.

The locked stub code in the editor reads the input and tests the correctness of the Student and Priorities classes implementation.

Constraints

```
1 import java.util.ArrayList;
2 import java.util.List;
3 import java.util.Scanner;
4
5 /*
6  * Create the Student and Priorities classes here.
7  */import java.util.PriorityQueue;
8 import java.util.Comparator;
9
10 class Student {
11     private final int id;
12     private final String name;
13     private final double cgpa;
14
15     public Student(int id, String name, double cgpa) {
16         this.id = id;
17         this.name = name;
18         this.cgpa = cgpa;
19     }
20
21     public int getID() {
22         return id;
23     }
24
25     public String getName() {
26         return name;
27     }
28
29     public double getCGPA() {
30         return cgpa;
31     }
32 }
33
34 class Priorities {
35     public List<Student> getStudents(List<String> events) {
36         PriorityQueue<Student> pq = new PriorityQueue<>(
37             Comparator.comparing(Student::getCGPA).reversed()
38                 .thenComparing(Student::getName)
39                 .thenComparing(Student::getID)
40         );
41
42         for (String event : events) {
43             String[] parts = event.split(" ");
44             // ... (rest of the code) ...
45         }
46     }
47 }
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 12
2 ENTER John 3.75 50
3 ENTER Mark 3.8 24
4 ENTER Shafaet 3.7 35
5 SERVED
6 SERVED
7 ENTER Samiha 3.85 36
8 SERVED
9 ENTER Ashley 3.9 42
10 ENTER Maria 3.6 46
11 ENTER Anik 3.95 49
```

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Java Arraylist

[Problem](#)[Submissions](#)[Leaderboard](#)[Discussions](#)[Editorial !\[\]\(9479d69b60a82161c6862eaa53eb4db3_img.jpg\)](#)

Sometimes it's better to use dynamic size arrays. Java's [Arraylist](#) can provide you this feature. Try to solve this problem using Arraylist.

You are given n lines. In each line there are zero or more integers. You need to answer a few queries where you need to tell the number located in y^{th} position of x^{th} line.

Take your input from System.in.

Input Format

The first line has an integer n . In each of the next n lines there will be an integer d denoting number of integers on that line and then there will be d space-separated integers. In the next line there will be an integer q denoting number of queries. Each query will consist of two integers x and y .

Constraints

- $1 \leq n \leq 20000$
- $0 \leq d \leq 50000$
- $1 \leq q \leq 1000$
- $1 \leq x \leq n$

Each number will fit in signed integer.

Total number of integers in n lines will not cross 10^5 .

Output Format

In each line, output the number located in y^{th} position of x^{th} line. If there is no such position, just print "ERROR!"

Sample Input

```
5
5 41 77 74 22 44
1 12
4 37 34 36 52
0
3 20 22 33
5
1 3
3 4
3 1
```

```

1  import java.io.*;
2  import java.util.*;
3
4  public class Solution {
5
6      public static void main(String[] args) {
7          Scanner scan = new Scanner(System.in);
8
9          // Read the number of lines
10         int n = scan.nextInt();
11
12         // Create a 2D ArrayList (a list containing lists of integers)
13         List<List<Integer>> lines = new ArrayList<>();
14
15         for (int i = 0; i < n; i++) {
16             int d = scan.nextInt(); // Number of integers on this specific line
17             List<Integer> currentLine = new ArrayList<>();
18             for (int j = 0; j < d; j++) {
19                 currentLine.add(scan.nextInt());
20             }
21             lines.add(currentLine);
22         }
23
24         // Read the number of queries
25         int q = scan.nextInt();
26
27         for (int i = 0; i < q; i++) {
28             int x = scan.nextInt(); // Line number (1-indexed)
29             int y = scan.nextInt(); // Position number (1-indexed)
30
31             // Convert 1-based indexing from input to 0-based indexing for Java ArrayLists
32             int lineIdx = x - 1;
33             int posIdx = y - 1;
34
35             // Check if the requested line exists and if the position exists within that line
36             if (lineIdx >= 0 && lineIdx < lines.size() && posIdx >= 0 && posIdx < lines.get(lineIdx).size()) {
37                 System.out.println(lines.get(lineIdx).get(posIdx));
38             } else {
39                 System.out.println("ERROR!");
40             }
41         }
42
43         scan.close();

```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

Input (stdin)

Download

```
1 5
2 5 41 77 74 22 44
3 1 12
4 4 37 34 36 52
5 0
6 3 20 22 33
7 5
8 1 3
9 3 4
10 3 1
11 4 3
```

Problem List

179. Largest Number

Medium

Topics

Companies

Description

Editorial

Solutions

Accepted

Submissions

Solved

Given a list of non-negative integers `nums`, arrange them such that they form the largest number and return it.

Since the result may be very large, so you need to return a string instead of an integer.

Example 1:

Input: `nums = [10,2]`
Output: `"210"`

Example 2:

Input: `nums = [3,30,34,5,9]`
Output: `"9534330"`

Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 10^9$

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Seen this question in a real interview before?

1/6

Yes

No

Accepted 862,419/2M

Acceptance Rate 43.5%

Topics

9.5K

228

59 Online

Code

Java

Auto

```
9 Arrays.sort(arr, (a, b) -> (b + a).compareTo(a + b));
10
11 // If the largest element is "0", all elements are 0
12 if (arr[0].equals("0")) {
13     return "0";
14 }
15
16 StringBuilder sb = new StringBuilder();
17
18 for (String s : arr) {
19     sb.append(s);
20 }
21
22 return sb.toString();
23 }
24 }
```

Saved

Ln 24, Col 2

Testcase

Test Result

Accepted

Runtime: 1 ms

Case 1

Case 2

Input

nums =
[10,2]

Output

"210"

Expected

"210"

Contribute a testcase

2418. Sort the People

Easy Topics Companies Hint

You are given an array of strings `names`, and an array `heights` that consists of **distinct** positive integers. Both arrays are of length `n`.

For each index `i`, `names[i]` and `heights[i]` denote the name and height of the `i`th person.

Return `names` sorted in **descending** order by the people's heights.

Example 1:

Input: `names = ["Mary","John","Emma"], heights = [180,165,170]`

Output: `["Mary","Emma","John"]`

Explanation: Mary is the tallest, followed by Emma and John.

Example 2:

Input: `names = ["Alice","Bob","Bob"], heights = [155,185,150]`

Output: `["Bob","Alice","Bob"]`

Explanation: The first Bob is the tallest, followed by Alice and the second Bob.

Constraints:

- `n == names.length == heights.length`
- `1 <= n <= 103`
- `1 <= names[i].length <= 20`
- `1 <= heights[i] <= 105`
- `names[i]` consists of lower and upper case English letters.
- All the values of `heights` are distinct.

1.9K 180 10 Online

Java Auto

```
10         map.put(heights[i], names[i]);
11     }
12
13     // Sort heights in ascending order
14     Arrays.sort(heights);
15
16     // Populate the result array in descending order (tallest to shortest)
17     String[] result = new String[n];
18     for (int i = 0; i < n; i++) {
19         result[i] = map.get(heights[n - 1 - i]);
20     }
21
22     return result;
23 }
24 }
25
```

Saved

Ln 25, Col 1

Testcase Test Result

Accepted Runtime: 1 ms

Case 1 Case 2

Input

names =
["Mary", "John", "Emma"]

heights =
[180, 165, 170]

Output

["Mary", "Emma", "John"]

Expected

["Mary", "Emma", "John"]

Java Comparator

[Problem](#)[Submissions](#)[Leaderboard](#)[Discussions](#)[Editorial !\[\]\(0230214116c86dbf511158ea2e1aae13_img.jpg\)](#)

Comparators are used to compare two objects. In this challenge, you'll create a comparator and use it to sort an array.

The `Player` class is provided for you in your editor. It has **2** fields: a ***name*** String and a ***score*** integer.

Given an array of ***n*** `Player` objects, write a comparator that sorts them in order of decreasing score; if **2** or more players have the same score, sort those players alphabetically by name. To do this, you must create a `Checker` class that implements the `Comparator` interface, then write an `int compare(Player a, Player b)` method implementing the `Comparator.compare(T o1, T o2)` method.

Input Format

Input from stdin is handled by the locked stub code in the `Solution` class.

The first line contains an integer, ***n***, denoting the number of players.

Each of the ***n*** subsequent lines contains a player's ***name*** and ***score***, respectively.

Constraints

- $0 \leq \text{score} \leq 1000$
- 2 players can have the same name.
- Player names consist of lowercase English letters.

Output Format

You are not responsible for printing any output to stdout. The locked stub code in `Solution` will create a `Checker` object, use it to sort the `Player` array, and print each sorted element.

Sample Input

```
5
amy 100
david 100
heraldo 50
aakansha 75
aleksa 150
```

Change Theme

Language

Java 8



```
1 import java.util.*;
2
3 class Checker implements Comparator<Player> {
4     @Override
5     public int compare(Player a, Player b) {
6         // Rule 1: Sort by decreasing score
7         if (a.score != b.score) {
8             return Integer.compare(b.score, a.score);
9         }
10        // Rule 2: If scores are equal, sort alphabetically by name (ascending)
11        return a.name.compareTo(b.name);
12    }
13 }
14 // Write your Checker class here
15
16 > class Player{ ...
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 5
2 amy 100
3 david 100
4 heraldo 50
5 aakansha 75
6 aleksa 150
```

Your Output (stdout)

```
1 aleksa 150
2 amy 100
3 david 100
```

[Download](#)